

KONGU ENGINEERING COLLEGE
(AUTONOMOUS)

PERUNDURAI, ERODE – 638 060

EXPERIMENT - 12

PROJECT TITLE

**AUTOMATED COLOUR BASED PRODUCT
SORTING SYSTEM**

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AUTOMATED COLOUR BASED PRODUCT SORTING SYSTEM

1. Project review:

The **Automated Colour Based Product Sorting System** is designed to automatically identify and sort products according to their colour. The system uses an **ESP8266 NodeMCU** as the main controller and a **TCS3200 colour sensor** to detect the colour of the product.

The colour sensor detects the red, green, and blue colour components of the object. The ESP8266 processes the sensor data and identifies the product colour. Based on the detected colour, a **servo motor** moves the product into the corresponding collection area.

This system reduces manual sorting, improves speed, and provides an efficient method for industrial automation.

2. Problem Statement:

In industries, products are often sorted manually based on colour and other properties. Manual sorting requires more time and labour and may result in human errors.

Therefore, an automated system is required to identify the colour of products and sort them into appropriate categories.

The proposed system uses a **colour sensor, ESP8266 NodeMCU, servo motor, buzzer, and LED** to automatically detect and sort products according to their colour.

3. Proposed Solution:

The proposed system uses a **TCS3200 colour sensor** to detect the colour of a product.

The ESP8266 receives the sensor readings and processes the colour information. According to the detected colour, the servo motor rotates to a predefined position and directs the product into the appropriate sorting section.

Proposed Operation:

Start the system and initialize the components.

Detect the product using the colour sensor.

Read the colour values.

Process the sensor readings.

Identify the product colour.

Compare the detected colour with predefined colour ranges.

Rotate the servo motor according to the detected colour.

Sort the product into the correct section.

Provide an indication using LED or buzzer.

Repeat the sorting process continuously.

4. System Architecture:

The Automated Colour Based Product Sorting System consists of the following sections:

Input Section:

TCS3200 Colour Sensor – Detects the colour of the product.

Processing Section:

The ESP8266 NodeMCU acts as the central controller.

- Receives colour sensor data.
- Processes RGB colour values.
- Identifies the product colour.
- Controls the servo motor and indication devices.

Sorting Section

- **Servo Motor** – Moves the sorting mechanism according to the detected colour.

Example sorting positions:

- Red Product → Left Section

- Green Product → Centre Section
- Blue Product → Right Section

Indication Section:

- **LED** – Provides visual indication.
- **Buzzer** – Provides an audible indication when a product is detected and sorted.

5. Circuit Table:

Important Pin Mapping:

ESP8266 NodeMCU Pin	Component
D1	GPIO5 TCS3200 → S0
D2	GPIO4 TCS3200 → S1
D5	GPIO14 TCS3200 → S2
D6	GPIO12 TCS3200 → S3
D7	GPIO13 TCS3200 → OUT
D4	GPIO2 Servo Motor Signal
D3	GPIO0 LED
D8	GPIO15 Buzzer
3.3V / 5V	– Sensor Power
GND	– Common Ground

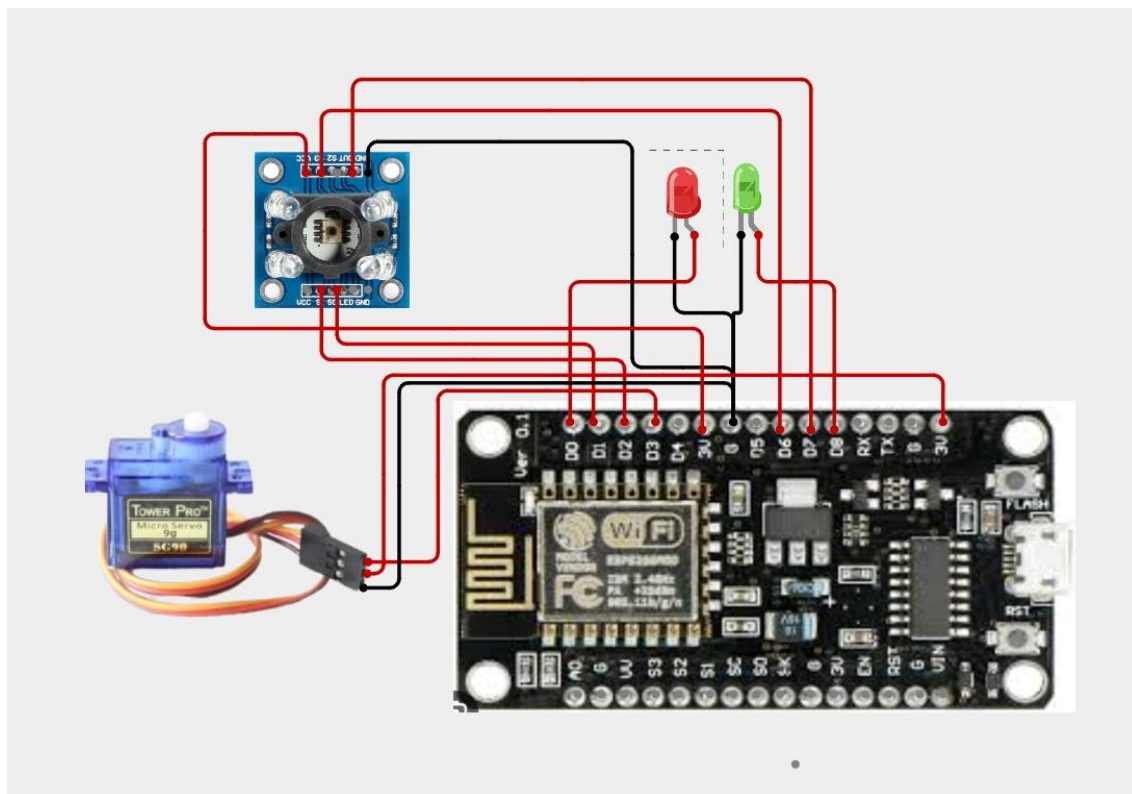
6. Circuit Design:

The circuit consists of an **ESP8266 NodeMCU**, **TCS3200 colour sensor**, **servo motor**, **LED**, **buzzer**, **breadboard**, and **jumper wires**.

The TCS3200 colour sensor is connected to the ESP8266 and used to detect the colour of the product. The sensor output is processed by the NodeMCU.

The servo motor is connected to the ESP8266 and controls the sorting mechanism. The LED and buzzer provide visual and audible indications.

When a product is placed in front of the sensor, its colour is detected. The ESP8266 identifies the colour and rotates the servo motor to the required position for sorting.



7. Algorithm:

Start the system.

Initialize the **ESP8266 NodeMCU**, **TCS3200 colour sensor**, **servo motor**, **LED**, and **buzzer**.

Detect the product placed in front of the colour sensor.

Read the **Red, Green, and Blue (RGB)** colour values.

Process and compare the RGB values to identify the product colour.

Check whether the detected colour is **Red, Green, or Blue**.

Rotate the **servo motor** to the corresponding sorting position.

Sort the product into the appropriate collection section.

Turn **ON the LED and buzzer briefly** to indicate successful sorting.

Wait for the next product and **repeat the sorting process continuously**.

8. Coding:

```
#include <Servo.h>
```

```
// =====
```

```
// AUTOMATED COLOR-BASED PRODUCT SORTING SYSTEM
```

```
// ESP8266 NodeMCU
```

```
// TCS3200 + Servo + 2 LEDs
```

```
// =====
```

```
// =====
```

```
// TCS3200 CONNECTIONS
```

```
// =====
```

```
#define S0_PIN 5    // D1 / GPIO5
```

```
#define S1_PIN 4    // D2 / GPIO4
```

```
#define S2_PIN 14   // D5 / GPIO14
```

```
#define S3_PIN 12   // D6 / GPIO12
```

```
#define COLOR_OUT 13 // D7 / GPIO13
```

```
// =====
```

```
// SERVO
```

```
// =====
```

```
#define SERVO_PIN 0 // D3 / GPIO0
```

```
// =====
```

```
// LEDS
```

```
// =====
```

```
#define RED_LED 16  // D0 / GPIO16
```

```
#define GREEN_LED 15 // D8 / GPIO15
```

```
// =====
```

```
// SERVO OBJECT
```

```
// =====
```

Servo sortingServo;

// =====

// VARIABLES

// =====

unsigned long redFrequency;

unsigned long greenFrequency;

unsigned long blueFrequency;

// =====

// SETUP

// =====

void setup()

{

Serial.begin(9600);

delay(1000);

// -----

// TCS3200 pins

// -----


```
pinMode(S0_PIN, OUTPUT);
```

```
pinMode(S1_PIN, OUTPUT);
```

```
pinMode(S2_PIN, OUTPUT);
```

```
pinMode(S3_PIN, OUTPUT);
```

```
pinMode(COLOR_OUT, INPUT);
```

```
// -----
```

```
// LED pins
```

```
// -----
```

```
pinMode(RED_LED, OUTPUT);
```

```
pinMode(GREEN_LED, OUTPUT);
```

```
digitalWrite(RED_LED, LOW);
```

```
digitalWrite(GREEN_LED, LOW);
```

```
// -----
```

```
// TCS3200 frequency scaling
```

```
// 20% scaling
```

```
// -----
```

```
digitalWrite(S0_PIN, HIGH);
```

```
digitalWrite(S1_PIN, LOW);
```

```
// -----  
// Servo  
// -----
```

```
sortingServo.attach(SERVO_PIN);
```

```
sortingServo.write(90);
```

```
delay(1000);
```

```
// -----  
// Startup message  
// -----
```

```
Serial.println();
```

```
Serial.println("=====");
```

```
Serial.println(" AUTOMATED COLOR SORTING SYSTEM");
```

```
Serial.println(" ESP8266 NODEMCU");
```

```
Serial.println(" TCS3200 + SERVO");
```

```
Serial.println("=====");
```

```
Serial.println("System Ready");
```

```
delay(2000);
```

```
}
```

```
// =====
```

```
// READ RED FREQUENCY
```

```
// =====
```

```
unsigned long readRed()
```

```
{
```

```
    digitalWrite(S2_PIN, LOW);
```

```
    digitalWrite(S3_PIN, LOW);
```

```
    delay(50);
```

```
    unsigned long frequency =
```

```
        pulseIn(
```

```
            COLOR_OUT,
```

```
            LOW,
```

```
            100000
```

```
        );
```

```
    return frequency;
```

```
}
```

```
// =====
```

```
// READ GREEN FREQUENCY
```

```
// =====
```

```
unsigned long readGreen()
{
    digitalWrite(S2_PIN, HIGH);
    digitalWrite(S3_PIN, HIGH);

    delay(50);

    unsigned long frequency =
        pulseIn(
            COLOR_OUT,
            LOW,
            100000
        );

    return frequency;
}
```

```
// =====
```

```
// READ BLUE FREQUENCY
```

```
// =====
```

```
unsigned long readBlue()
{
    digitalWrite(S2_PIN, LOW);
```

```
digitalWrite(S3_PIN, HIGH);
```

```
delay(50);
```

```
unsigned long frequency =
```

```
  pulseIn(
```

```
    COLOR_OUT,
```

```
    LOW,
```

```
    100000
```

```
  );
```

```
return frequency;
```

```
}
```

```
//=====
```

```
// RED PRODUCT SORTING
```

```
//=====
```

```
void sortRed()
```

```
{
```

```
  Serial.println();
```

```
  Serial.println("=====");
```

```
  Serial.println("COLOR DETECTED : RED");
```

```
  Serial.println("SORTING    : RED BIN");
```

```
  Serial.println("SERVO ANGLE  : 40 DEG");
```

```
  Serial.println("=====");
```

```
// Red LED ON
```

```
digitalWrite(  
  RED_LED,  
  HIGH  
);
```

```
// Green LED OFF
```

```
digitalWrite(  
  GREEN_LED,  
  LOW  
);
```

```
// Move sorting arm
```

```
sortingServo.write(40);
```

```
delay(1000);
```

```
// Return to center
```

```
sortingServo.write(90);
```

```
delay(500);
```

```
// Turn LED OFF
```

```
digitalWrite(  
  RED_LED,  
  LOW  
);  
}
```

```
// =====  
// GREEN PRODUCT SORTING  
// =====
```

```
void sortGreen()  
{  
  Serial.println();  
  Serial.println("=====");  
  Serial.println("COLOR DETECTED : GREEN");  
  Serial.println("SORTING      : GREEN BIN");  
  Serial.println("SERVO ANGLE   : 140 DEG");  
  Serial.println("=====");  
}
```

```
// Green LED ON
```

```
digitalWrite(  
  GREEN_LED,  
  HIGH  
);
```

```
// Red LED OFF
```

```
digitalWrite(  
  RED_LED,  
  LOW  
);
```

```
// Move sorting arm  
sortingServo.write(140);
```

```
delay(1000);
```

```
// Return to center  
sortingServo.write(90);
```

```
delay(500);
```

```
// Turn LED OFF
```

```
digitalWrite(  
  GREEN_LED,  
  LOW  
);
```

```
}
```

```
// =====
```



```
// UNKNOWN PRODUCT
```

```
//=====
```

```
void unknownProduct()
```

```
{
```

```
  Serial.println();
```

```
  Serial.println("=====");
```

```
  Serial.println("COLOR DETECTED : UNKNOWN");
```

```
  Serial.println("SORTING      : NO ACTION");
```

```
  Serial.println("=====");
```

```
  digitalWrite(
```

```
    RED_LED,
```

```
    LOW
```

```
  );
```

```
  digitalWrite(
```

```
    GREEN_LED,
```

```
    LOW
```

```
  );
```

```
  sortingServo.write(90);
```

```
  delay(500);
```

```
}
```

```
// =====  
// MAIN LOOP  
// =====
```

```
void loop()
```

```
{
```

```
  // -----
```

```
  // Read color frequencies
```

```
  // -----
```

```
  redFrequency = readRed();
```

```
  greenFrequency = readGreen();
```

```
  blueFrequency = readBlue();
```

```
  // -----
```

```
  // Display sensor readings
```

```
  // -----
```

```
  Serial.println();
```

```
  Serial.println("-----");
```

```
  Serial.print(
```

```
    "RED Frequency  : "
```

```
  );
```

```
Serial.println(  
    redFrequency  
);
```

```
Serial.print(  
    "GREEN Frequency : "  
);
```

```
Serial.println(  
    greenFrequency  
);
```

```
Serial.print(  
    "BLUE Frequency : "  
);
```

```
Serial.println(  
    blueFrequency  
);
```

```
// -----  
// Color detection  
// -----
```

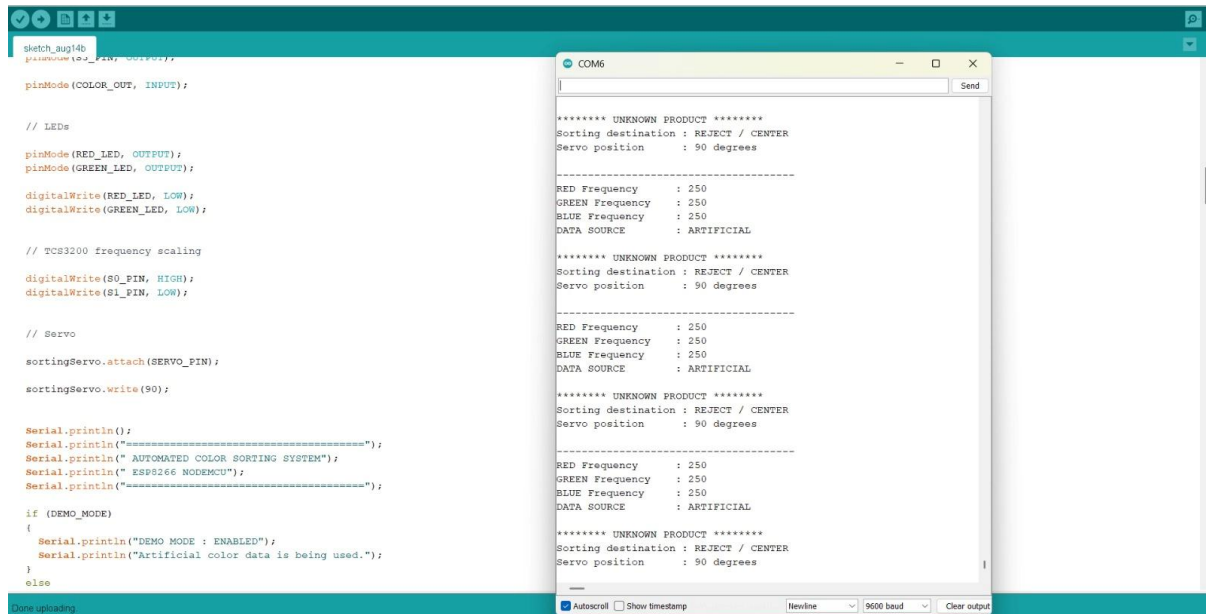
```
if (  
    redFrequency > 0 &&  
    redFrequency < greenFrequency &&  
    redFrequency < blueFrequency  
)  
{  
    sortRed();  
}
```

```
else if (  
    greenFrequency > 0 &&  
    greenFrequency < redFrequency &&  
    greenFrequency < blueFrequency  
)  
{  
    sortGreen();  
}
```

```
else  
{  
    unknownProduct();  
}
```

```
delay(1000);  
}
```

Coding Output:



The screenshot shows the Arduino IDE interface. On the left, the sketch editor displays the following code:

```
sketch_aug14b
#include <Servo.h>

pinMode(COLOR_OUT, INPUT);

// LEDs
pinMode(RED_LED, OUTPUT);
pinMode(GREEN_LED, OUTPUT);

digitalWrite(RED_LED, LOW);
digitalWrite(GREEN_LED, LOW);

// TCS3200 frequency scaling
digitalWrite(S0_PIN, HIGH);
digitalWrite(S1_PIN, LOW);

// Servo
sortingServo.attach(SERVO_PIN);
sortingServo.write(90);

Serial.println();
Serial.println("=====");
Serial.println("  AUTOMATED COLOR SORTING SYSTEM");
Serial.println(" ESP8266 NODEMCU");
Serial.println("=====");

if (DEMO_MODE)
{
  Serial.println("DEMO MODE : ENABLED");
  Serial.println("Artificial color data is being used.");
}
else
```

On the right, the Serial Monitor window (COM6) shows the output of the sketch:

```
***** UNKNOWN PRODUCT *****
Sorting destination : REJECT / CENTER
Servo position      : 90 degrees

-----
RED Frequency       : 250
GREEN Frequency      : 250
BLUE Frequency       : 250
DATA SOURCE          : ARTIFICIAL

***** UNKNOWN PRODUCT *****
Sorting destination : REJECT / CENTER
Servo position      : 90 degrees

-----
RED Frequency       : 250
GREEN Frequency      : 250
BLUE Frequency       : 250
DATA SOURCE          : ARTIFICIAL

***** UNKNOWN PRODUCT *****
Sorting destination : REJECT / CENTER
Servo position      : 90 degrees

-----
RED Frequency       : 250
GREEN Frequency      : 250
BLUE Frequency       : 250
DATA SOURCE          : ARTIFICIAL

***** UNKNOWN PRODUCT *****
Sorting destination : REJECT / CENTER
Servo position      : 90 degrees
```

The Serial Monitor window also shows the status bar at the bottom: "Autoscroll" is checked, "Show timestamp" is unchecked, the baud rate is set to "9600 baud", and there is a "Clear output" button.

9. System Working:

Power ON the System:

The ESP8266 NodeMCU is powered ON.

Initialize Components:

The TCS3200 colour sensor, servo motor, LED, and buzzer are initialized.

Detect Product Colour

The product is placed in front of the colour sensor.

The TCS3200 detects the colour components.

Process Colour Data

The ESP8266 reads the red, green, and blue values.

Sorting Operation

Depending on the detected colour:

- Red → Servo moves to Left
- Green → Servo moves to Centre
- Blue → Servo moves to Right

Indication

LED → ON

Buzzer → ON briefly

Repeat the Process:

- The system continuously detects and sorts products according to their colour.

10. Prototype Implementation:

The prototype is implemented using an ESP8266 NodeMCU, TCS3200 colour sensor, servo motor, LED, buzzer, breadboard, and jumper wires.

Hardware Setup

Connect the TCS3200 sensor pins to the ESP8266.

Connect the servo motor signal pin to D4.

Connect the LED to D3 through a suitable resistor.

Connect the buzzer to D8.

Connect all GND terminals to a common ground.

Upload the program using the Arduino IDE.

Power ON the system.

Place a coloured product in front of the sensor.

Observe the detected RGB values.

Observe the servo motor movement.

The product is directed into the corresponding sorting section.

Repeat the process for different coloured products.

Prototype Working:

- The colour sensor detects the colour of the product.
- The ESP8266 processes the sensor output.
- The product colour is identified.
- The servo motor moves to the required sorting position.
- The LED and buzzer indicate the sorting operation.
- The system continuously performs automatic colour-based sorting.

11. Bill of Materials:

S.No.	Component	Specification	Quantity
1	ESP8266	NodeMCU Development Board	1
2	TCS3200	Colour Sensor Module	1
3	Servo Motor	SG90 / Suitable Servo Motor	1
4	LED	Standard Indicator LED	1
5	Buzzer	3.3V/5V Buzzer	1
6	Resistor	Suitable LED Current Limiting Resistor	1
7	Breadboard	Standard Prototype Board	1
8	Jumper Wires	Male-Male / Male-Female	1 Set
9	USB Cable	Micro-USB	1
10	Power Supply	Suitable Regulated Supply	1

12. Results & Performance Analysis:

The prototype successfully demonstrates automatic colour detection and product sorting.

- The TCS3200 sensor detects the colour of the product.

- The ESP8266 processes the colour sensor readings.
- The system identifies red, green, and blue products.
- The servo motor automatically moves to different positions.
- The products can be sorted into separate sections.
- The LED and buzzer provide sorting indications.

Results

The colour sensor successfully detects product colours.

The ESP8266 successfully processes the colour readings.

The system automatically identifies different colours.

Red, green, and blue products are sorted into separate sections.

The servo motor successfully controls the sorting mechanism.

The LED provides a visual indication.

The buzzer provides an audible indication.

The prototype demonstrates the basic operation of an **Automated Colour Based Product Sorting System**.

Performance Analysis:

Parameter	Expected Performance
Colour Detection	Automatic
Data Processing	Real-time
Product Sorting	Automatic
Red Product	Left Position
Green Product	Centre Position
Blue Product	Right Position
Visual Indication	LED ON
Audible Indication	Buzzer ON Briefly

